

1. Independence of Random Variables

Mutual independence: Random variables $R_1, R_2, \dots, R_n$ are mutually independent if for all $x_1 \in \text{Range}(R_1), x_2 \in \text{Range}(R_2), \dots, x_n \in \text{Range}(R_n)$, events $[R_1 = x_1], [R_2 = x_2], \dots, [R_n = x_n]$ are mutually independent.

Mutual independence of events: A set of events is said to be mutually independent if the probability of each event in the set is the same no matter which subset of events has occurred:

- Formally, $E_1, E_2, \dots, E_n$ are mutually independent if: $\forall i \in [n], \Pr[E_i \mid \text{any subset of the other } n-1 \text{ events}] = \Pr[E_i]$

Example: For events E_1, E_2, E_3 to be mutually independent, the following equalities should hold:

- $\forall i, j \neq i, \Pr[E_i \mid E_j] = \Pr[E_i]$
- $\forall i, \forall j, k, j \neq i, k \neq i, j \neq k, \Pr[E_i \mid E_j \cap E_k] = \Pr[E_i]$

- For 2 r.v's R_1, R_2 , mutual independence and pairwise independence are equivalent.
- For more than 2 r.v's, mutual independence implies pairwise independence.

2. Max Cut

Aim: Given a graph $G = (\mathcal{V}, \mathcal{E})$, partition the graph's vertices into two complementary sets $\mathcal{S}$ and $\mathcal{T}$, such that the number of edges between the set $\mathcal{S}$ and the set $\mathcal{T}$ is as large as possible.

- Max Cut has applications to VLSI circuit design.

Formal objective: Find a set $\mathcal{U} \subseteq \mathcal{V}$ of vertices that solve the following problem:

$$\max_{\mathcal{U} \subseteq \mathcal{V}} |\delta(\mathcal{U})| \text{ where } \delta(\mathcal{U}) := \{(u, v) \in \mathcal{E} \mid (u \in \mathcal{U} \text{ and } v \notin \mathcal{U}) \text{ or } (u \notin \mathcal{U} \text{ and } v \in \mathcal{U})\}$$

$\delta(\mathcal{U})$ is the “cut” corresponding to the set $\mathcal{U}$ and the aim is to find the cut with the largest size.

- Max Cut is **NP-hard**, meaning that there is no polynomial (in $|\mathcal{E}|$) time algorithm that solves Max Cut exactly.
- We want to find an approximate solution $\mathcal{U}$ such that, if OPT is the size of the optimal cut, then, $|\delta(\mathcal{U})| \geq \alpha \text{ OPT}$ where $\alpha \in (0, 1)$ is the multiplicative approximation factor.
- Randomized algorithm that guarantees an approximate solution with $\alpha = \frac{1}{2}$ with probability close to 1 (Erdos, 1967).
- Complicated algorithm with $\alpha = 0.878$ (Goemans and Williamson, 1995).
- Under some technical conditions, no algorithm has $\alpha > 0.878$ (Khot et al, 2004).

We will use Erdos' randomized algorithm and prove the result in expectation. We wish to prove that for $\mathcal{U}$ returned by Erdos' algorithm,

$$\mathbb{E}[|\delta(\mathcal{U})|] \geq \frac{1}{2} \text{ OPT}$$

Algorithm: Select $\mathcal{U}$ to be a random subset of $\mathcal{V}$ i.e. for each vertex v , choose v to be in the set $\mathcal{U}$ independently with probability $\frac{1}{2}$ (don't even look at the edges).

2.1. Deviation from the Mean

- We have developed tools to calculate the mean of random variables. Getting a handle on the expectation is useful because it tells us what would happen on average.
- However, summarizing the PDF using the mean is typically not enough. We also want to know how “spread” the distribution is.

Example: Consider three random variables W, Y, Z whose PDFs can be given as:

$$\begin{aligned}
W &= 0 \text{ with } p = 1 \\
Y &= \begin{cases} -1 & \text{with } p = \frac{1}{2} \\ +1 & \text{with } p = \frac{1}{2} \end{cases} \\
Z &= \begin{cases} -1000 & \text{with } p = \frac{1}{2} \\ +1000 & \text{with } p = \frac{1}{2} \end{cases}
\end{aligned}$$

Though $\mathbb{E}[W] = \mathbb{E}[Y] = \mathbb{E}[Z] = 0$, these distributions are quite different. Z can take values really far away from its expected value, while W can only take one value equal to the mean. Hence, we want to understand how much a random variable **deviates** from its mean.

Before we calculate the deviation of a r.v. from its mean, we need an additional definition: For an r.v. $X : \mathcal{S} \rightarrow \mathcal{V}$, and a function $g : \mathcal{V} \rightarrow \mathbb{R}$, we define $\mathbb{E}[g(X)]$ as follows:

$$\mathbb{E}[g(X)] = \sum_{x \in \text{Range}(X)} g(x) * \mathbb{E}[X]$$

If $g(x) = x$ for all $x \in \text{Range}(X)$, then $\mathbb{E}[g(X)] = \mathbb{E}[X]$.

Q: For a standard dice, if X is the r.v. corresponding to the number that comes up on the dice, compute $\mathbb{E}[X^2]$ and $\mathbb{E}[X]$.

For a standard dice, $X \sim \text{Uniform}(\{1, 2, 3, 4, 5, 6\})$ and hence,

$$\mathbb{E}[X^2] = \sum_{x \in \{1, 2, 3, 4, 5, 6\}} x^2 \Pr[X = x] = \frac{1}{6} [1^2 + \dots + 6^2] = \frac{91}{6}$$

and

$$(\mathbb{E}[X])^2 = \left(\sum_{x \in \{1, 2, 3, 4, 5, 6\}} x \Pr[X = x] \right)^2 = \left(\frac{1}{6} [1 + \dots + 6] \right)^2 = \frac{49}{4}$$

3. Variance

Definition: *Variance* is the standard way to measure the deviation of a r.v. from its mean. Formally, for an r.v. X ,

$$\text{Var}[X] = \mathbb{E}[(X - \mathbb{E}[X])^2] = \sum_{x \in \text{Range}(X)} (x - \mu)^2 \Pr[X = x] \quad (\text{where } \mu := \mathbb{E}[X])$$

Intuitively, the variance measures the probability-weighted average of how far (in squared distance) the random variable is from its mean μ .

Q: If $X \sim \text{Ber}(p)$, compute $\text{Var}[X]$.

Since X is a Bernoulli random variable, $X = 1$ with probability p and $X = 0$ with probability $1 - p$. Recall that $\mathbb{E}[X] = \mu = (0)(1 - p) + (1)(p) = p$.

$$\begin{aligned}
\text{Var}[X] &= \sum_{x \in \{0, 1\}} (x - p)^2 \Pr[X = x] = (0 - p)^2 \Pr[X = 0] + (1 - p)^2 \Pr[X = 1] \\
&= p^2(1 - p) + (1 - p)^2 p = p(1 - p)[p + 1 - p] = p(1 - p)
\end{aligned}$$

- For a Bernoulli r.v. X , $\text{Var}[X] = p(1 - p) \leq \frac{1}{4}$ and the variance is maximum when $p = \frac{1}{2}$.

Alternative definition of variance: $\text{Var}[X] = \mathbb{E}[X^2] - \mu^2 = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$